Lecture 4:

The chain Rule

~~Th~~ Here we will ~~menti~~ derive a formula that express the derivative of a composition $f \circ g$ in terms of the derivatives of f and g. This formula will enable us to differentiate complicated functions.

Theorem (Chain Rule):

If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f \circ g$ is differentiable at x. Moreover, if $y = f(g(x))$ and $u = g(x)$ then $y = f(u)$ and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

So, we can write alternate way,

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x).$$

e.g. find $\frac{dy}{dx}$ where $y = \tan(4x^3+4x)$.

$$\frac{dy}{dx} = \frac{d}{dx}\left[\tan(4x^3+4x)\right]$$

$$= \sec^2(4x^3+4x) \cdot \frac{d}{dx}(4x^3+4x)$$

$$= \sec^2(4x^3+4x) \cdot (12x^2+4)$$

$$= (12x^2+4)\sec^2(4x^3+4x)$$

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Example: ~~Find $\frac{dy}{dx}$ if $y=$~~

Find $\frac{d}{dx}\sin(\sqrt{1+\cos x})$.

$u=\sqrt{1+\cos x}$

$\frac{du}{dx} = \frac{-\sin x}{2\sqrt{1+\cos x}}$

$$\frac{d}{dx}\left(\sin\sqrt{1+\cos x}\right) = \frac{d}{dx}\sin u$$

$$= \cancel{\frac{d}{dx}(\sin u)}$$

$$= \cos u \frac{du}{dx}$$

$$= \cos\sqrt{1+\cos x}\cdot\frac{-\sin x}{2\sqrt{1+\cos x}}.$$

$$= -\frac{\sin x\cos\sqrt{1+\cos x}}{2\sqrt{1+\cos x}}$$

Example: Given that $f'(x)=\sqrt{3x+4}$ and $g(x)=x^2-1$, find $F'(x)$ if $F(x)=f(g(x))$.

Sol^n: Since

$$\frac{d}{dx}F(x) = f'(g(x))\cdot g'(x)$$

$f'(x)=\sqrt{3x+4}$. and $g(x)=x^2-1$ so $g'(x)=2x$

$$\therefore\ F'(x) = 2x\sqrt{3x^2+4}.$$

Note: If $f(x)=\sqrt{3x+4}$, $g(x)=x^2+1$, find $F'(x)$ where $F(x)=f(g(x))$.

Ans: $F'(x)=\left[f(g(x))\right]' = f'(g(x))\,g'(x)$ = Ans: $F'(x)=\frac{3x}{\sqrt{3x^2+1}}$

what about if $F(x)=g(f(x))$?

Extra PROBLEMS: page: 179

problem: 7–26, 76, 77, 78, 27, 31, 35, 36,

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Implicit Function

Definition: A given equation in x and y defines the function f is called implicitly if the graph of $y = f(x)$ coincides with a portion of the graph of the equation.

Example: Use implicit differentiation to find $\frac{dy}{dx}$ if

$$5y^2 + \sin y = x^2.$$

Sol^n

$$\frac{d}{dx}\left[5y^2 + \sin y\right] = \frac{d}{dx}(x^2)$$

$$5\frac{d}{dx}y^2 + \frac{d}{dx}\sin y = 2x$$

$$5 \cdot 2y\frac{dy}{dx} + \cos y\frac{dy}{dx} = 2x$$

$$10y\frac{dy}{dx} + \cos y\frac{dy}{dx} = 2x$$

$$\frac{dy}{dx}(10y + \cos y) = 2x$$

$$\frac{dy}{dx} = \frac{2x}{10y + \cos y}$$

Extra Problem:

Find $\frac{dy}{dx}$ by implicit differentiation.

1. $x^2y + 3xy^3 - x = 3$.
2. $\sin(x^2y^2) = x$.
3. $x^2 = \frac{x+y}{x-y}$.
4. $x^3 + y^3 = 3xy$.
5. $y + \sin y = x$.

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Logarithmic, ~~and~~ Exponential and Inverse Trigonometric Functions

Logarithmic

Formula: $f(x) = \ln x$ $\qquad \log_e x = \ln x$

So, (I) $f'(x) = \frac{d}{dx}[\ln(x)] = \frac{1}{x}$, $x > 0$

(II) $\frac{d}{dx}[\log_b x] = \frac{1}{\ln b}\frac{d}{dx}[\ln x]$ $\qquad \log_b x = \frac{\log_e x}{\log_e b} = \frac{\ln x}{\ln b}$

$= \frac{1}{x\ln(b)}$, $x > 0$

Example: Find $\frac{d}{dx}[\ln(x^2+1)]$.

Sol^n $\frac{d}{dx}[\ln(x^2+1)] = \frac{1}{x^2+1}\frac{d}{dx}(x^2+1)$

$= \frac{2x}{x^2+1}$ ※

Example: $\frac{d}{dx}\left[\ln\left(\frac{x^2\sin x}{\sqrt{1+x}}\right)\right] = \frac{d}{dx}\left[\ln x^2 + \ln\sin x - \ln(\sqrt{1+x})\right]$

$= \frac{d}{dx}\left[2\ln x + \ln\sin x - \frac{1}{2}\ln(1+x)\right]$

$= \frac{2}{x} + \frac{1}{\sin x}\cdot\cos x - \frac{1}{2}\frac{1}{1+x}$

$= \frac{2}{x} + \cot x - \frac{1}{2(1+x)}$

※

Extra problem

page: 198

Problems: 35-38

4, 6, 8, 15, 17, 16, 18

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Exponential Function

① $\frac{d}{dx}\left[e^{mx}\right] = e^{mx} \cdot \frac{d}{dx}(mx) = m e^{mx}$

② $\frac{d}{dx}\left[b^{x}\right] = b^{x} \ln b.$

Example:

1. $\frac{d}{dx}\left[2^{x}\right] = 2^{x} \ln(2).$

2. $\frac{d}{dx}\left[e^{x^3}\right] = e^{x^3} \frac{d}{dx}(x^3) = 3x^2 e^{x^3}$

Example: $\frac{d}{dx}(x^2+1)^{\sin x}$

Say, $y = (x^2+1)^{\sin x}$

$\ln y = \ln (x^2+1)^{\sin x} = \sin x \ln(x^2+1)$

$\frac{d}{dx}(\ln y) = \frac{d}{dx}\left[\sin x \ln (x^2+1)\right]$

$\frac{1}{y}\frac{dy}{dx} = \sin x \frac{d}{dx}\left[\ln(x^2+1)\right] + \ln(x^2+1) \frac{d}{dx}\left[\sin x\right]$

$= \sin x \cdot \frac{1}{x^2+1} \cdot 2x + \cos x \ln(x^2+1)$

$\frac{dy}{dx} = y\left[\frac{2x \sin x}{x^2+1} + \cos x \ln(x^2+1)\right]$

$= (x^2+1)^{\sin x}\left[\frac{2x \sin x}{x^2+1} + \cos x \ln(x^2+1)\right]$

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